

Analysis of the intra-aggregate pore structures in three soil types using X-ray computed tomography

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ABSTRACT

The pore structure of soils can control important physical and biological processes in soil-plant-microbial systems. However, quantifying the intra-aggregate pore structure is challenging. This study investigated the soil pore structure in aggregates from three typical zonal soil types (*alpine meadow* soil, *chestnut* soil and *red* soil) from cold, temperate and subtropical regions of China, respectively, using X-ray microscopic computed tomography (CT) with a resolution of 7 μm . The results indicated that the *chestnut* soil aggregates had higher porosity than the *alpine meadow* and *red* soil aggregates due to the presence of grass/shrub roots. The soil pores of the *chestnut* soil aggregates were mostly small, relatively regular and densely distributed. *Alpine meadow* soil aggregates had many large, elongated pores, which was attributed to the freeze-thaw cycles in alpine regions. The *alpine meadow* aggregate pores had the largest mean equivalent diameter, mean surface area and mean volume of the evaluated soil types. *Red* soil aggregates had low porosity and poor connectivity, which were attributed to the high temperatures and abundant precipitation in subtropical regions. Moreover, the soil aggregate pores of the three soil types were found to be dominated by micropores, especially the *chestnut* soil (90.39% of the number of pores). Soil texture significantly influences the total porosity of soil aggregates. Higher soil organic matter content may result in aggregate pores that are more irregular in shape. A soil pore architecture with high continuity likely facilitated greater water flow to the deeper soil layers in the *alpine meadow* soil than in the *red* and *chestnut* soils. However, in the *chestnut* soil aggregates, the water could be unavailable for uptake by plants due to the relatively poor soil pore structure. The results for the *red* soil aggregates are indicative of the poor structure and tendency toward rapid erosion of *red* soils.

1. Introduction

Soil structure, particularly the soil pore space arrangement, can control important physical and biological processes in soil-plant-microbial systems and can affect soil productivity and resistance to erosion and degradation (Six et al., 2000; Young and Crawford, 2004; Soinne et al., 2014; Ranjbar et al., 2016). The presence of pores can increase soil aeration, generate dominant flow or macropore flow, and promote the rapid movement of soil water (Lin et al., 2005; Jarvis, 2007). To date, extensive studies have investigated the influences of vegetation types, land use and anthropogenic disturbances on the 3D soil macropore structure at the scale of the soil column (Luo et al., 2010; Czachor et al., 2015; Hu et al., 2015; Bottinelli et al., 2016; Hu et al., 2018). For example, in China, Meng et al. (2017) quantified the soil macropore characteristics in different forest communities in a

mountainous area of North China. Hu et al. (2019) quantified the effects of enclosures on soil macropore and root characteristics in a shrub encroachment area. However, the quantification of soil pore structures remains challenging at the aggregate scale (Garbout et al., 2013).

X-ray computed tomography (CT), a nondestructive detection technology, has been used to quantify soil pore structure and provides a feasible, rapid, and accurate technical means for obtaining the three-dimensional structure of soil. Micro-CT provides fast scanning, strong imaging contrast, and high resolution, which enables the soil aggregate microstructures to be studied effectively (Dal Ferro et al., 2012; Czachor et al., 2015; Zhao et al., 2017a). Moreover, CT technology can not only observe the distribution of pore structures in two-dimensional images but also calculate the length, surface area and other geometric characteristics of the pores (Luo et al., 2008; Luo et al., 2010; Malik and Lu 2015; Zhao et al., 2017a). Galdos et al. (2019) used X-ray CT to

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Fig. 1. 3–5 mm aggregates for CT scans.

calculate and assess the shape, size and connectivity of pores in soil under zero-tillage and conventional tillage systems. Zhao et al. (2017a) quantified the structure of soil aggregates under four types of revegetation and a cropland on the Loess Plateau with synchrotron-based X-ray micro-CT. Thus, CT is an appropriate method for quantifying the soil aggregate structure.

Soil aggregates are indicators of soil structure (Wang et al., 2013; Ma et al., 2015; Menon et al., 2015) and are developed by the combination of mineral particles with substances such as roots, fungal hyphae, microbial polysaccharides and calcium bridges (Tisdall and Oades, 1982; Six et al., 2004; Bronick and Lal, 2005; Menon et al., 2015). The intra-aggregate spaces are relatively small and are the main driving factor influencing the heterogeneity of soil water transport spaces (Urbanek et al., 2007). Some previous studies have shown that the structure of soil aggregates is affected by soil texture and soil organic matter content (Henderson et al., 2005; De Jonge et al., 2009; Lugato et al., 2009; Zhao et al., 2010; Nakano et al., 2015; Zhao et al., 2017b). The structure and stability of aggregates affect the growth of crops by regulating water infiltration and movement, gas exchange, and biological activities (Bronick and Lal, 2005; Voltolini et al., 2017). However, the intra-aggregate pore structure is complex, and the pore structure of aggregates varies greatly among soil groups. As soil has a zonal distribution in space that is primarily determined by the interaction of climate and organisms (such as vegetation), we hypothesize that intra-aggregate pore structures vary significantly among different soil groups and that every soil group would have particular well-developed pore characteristics that are controlled by different biotic or abiotic factors. To verify this, we examined the pore structure of aggregates in three typical zonal soil groups from subtropical (*red soil*), temperate (*chestnut soil*) and cold (*alpine meadow soil*) regions of China using X-ray microscopic CT. The specific objectives of this study were to (1) visualize and quantify the structures of soil aggregates with X-ray CT and (2) ascertain the aggregate pore features and their controlling factors for the three typical zonal soil types.

2. Materials and methods

2.1. Study sites and sample preparation

Three soil types (*alpine meadow soil*, *red soil* and *chestnut soil*) were used in this experiment. These are typical zonal soils distributed in cold, temperate and subtropical regions of China, respectively. According to the USDA Soil Taxonomy, these three soil types are equivalent to Gleysols, Ultisols and Calcic-orthic Aridisols, respectively (Soil Survey Staff, 1987).

The *alpine meadow soil* was obtained from a pasture in Gangcha County, Qinghai Province, situated on the northeastern Qinghai-Tibet Plateau, with low temperature, aridity and frequent freeze-thaw cycles. Approximately 65% of the annual precipitation occurs in summer (An

et al., 2012). From 1959 to 2000, the average annual precipitation was 357 mm, and the evaporation was 925 mm (Li et al., 2007). The main plant species are *Kobresia pygmaea*, *Saussurea pulchra* and *Polygonum viviparum*, and the average vegetation cover is $97.00 \pm 2.00\%$.

The *red soil* was obtained from farmland in Nanchang, Jiangxi Province. The climate is a subtropical monsoon climate with a frost-free period of approximately 289 days. The average precipitation and evaporation are 1537 mm and 1150 mm, respectively. The average annual temperature is approximately 18.0 °C. The *chestnut soil* was obtained from a pasture in Taipusi County in the Inner Mongolia Autonomous Region of China. It is a typical grassland soil of Inner Mongolia, and it is representative of the Eurasian steppe region. The average annual temperature is 1.6 °C (Hu et al., 2018). The average annual precipitation in this region is 407 mm, and the average annual evaporation is 1750–2150 mm. The dominant plant species were *Caragana microphylla* L., *Stipa krylovii*, *Leymus chinensis*, and *Artemisia frigida* (Li et al. 2013).

Three plots with flat terrain were set up at each sampling site, and three samples were collected from each site. All soil types were sampled in July–August 2016. Topsoil samples (0–15 cm) were obtained from each point, and the samples from the three replicate points were mixed to obtain one composite sample for each plot. Soil aggregates of 3–5 mm were selected for analysis based on the instrument resolution and the sample size. Therefore, the soil samples were air-dried and dry-sieved to obtain 3–5 mm aggregates (Fig. 1). The soil particle size distribution was determined by the pipette method, and soil pH was determined using an acidometer (Sartorius PB-10 acidity meter) in a 1:2.5 ratio of soil to water. The soil organic carbon was measured by the method of Walkley and Black (1934) and converted to the organic matter content by multiplying by 1.724. The characteristics of the different soil types are listed in Table 1.

2.2. CT scanning and image analysis

A nanoVoxel-3502E X-ray three-dimensional microscopic CT (Sanying Precision Instruments Co., Ltd., China) was used to scan the soil aggregates with the X-ray source parameters of voltage 80 kV and current 50 μ A, with which 900 detailed and low-noise images could be obtained during a 360° rotation. The 3D reconstruction of aggregates was visualized by Avizo 9.0 software to quantify the microstructural

Table 1
Properties of the three soil types.

Sample	pH	Organic Matter (g kg ⁻¹)	Particle Size Composition (%)		
			Sand	Silt	Clay
<i>Alpine meadow soil</i>	7.89	4.14	46.12	30	25.88
<i>Red soil</i>	4.84	1.39	24.12	32	43.88
<i>Chestnut soil</i>	8.46	4.18	76.12	12	11.88

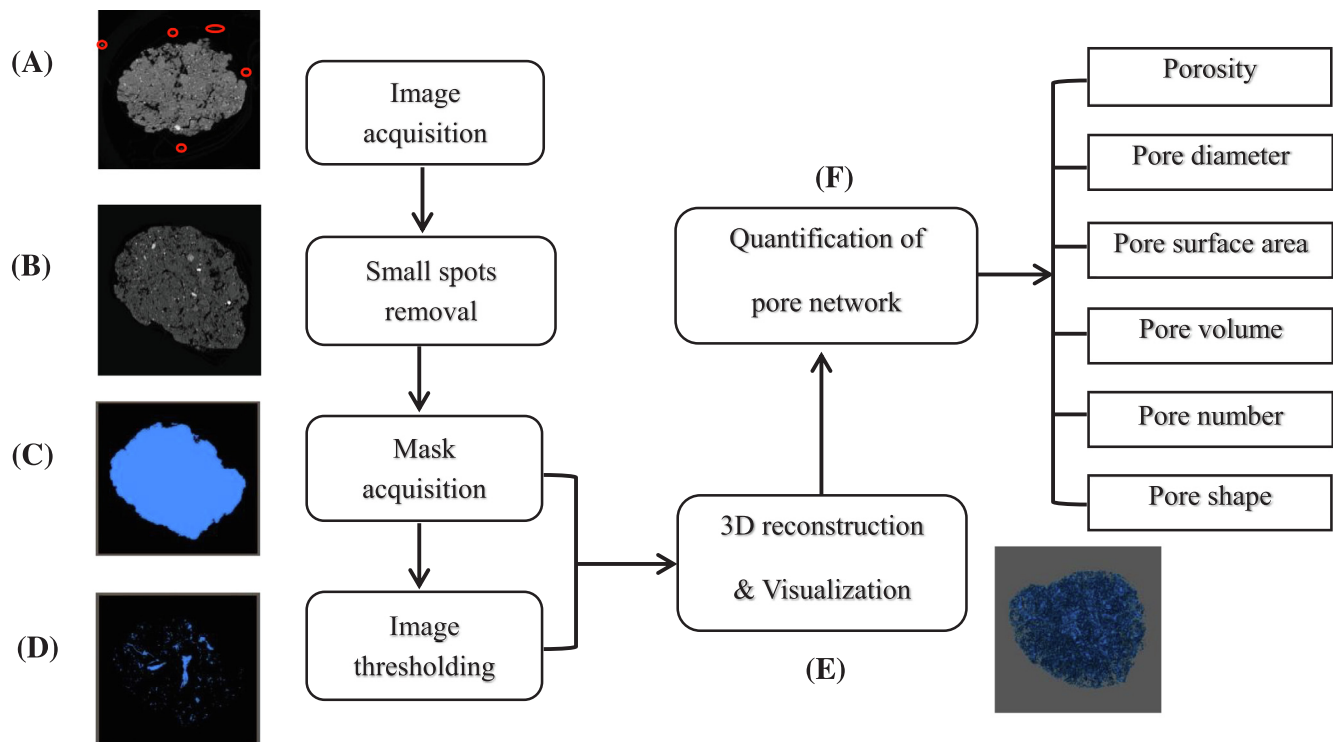


Fig. 2. Procedures used for the visualization and quantification of soil aggregate pore networks.

characteristics of different soil aggregates. The reconstructed images had $1700 \times 1700 \times 1500$ voxels, with a resolution of $7 \mu\text{m}$.

The CT-measured images showed the presence of small spots around the aggregates (Fig. 2-A). The spots were removed from the images by the volume editor module (Fig. 2-B). To analyze the soil intra-aggregate pore networks, the soil aggregate masks were obtained by selecting the soil matrix and then filling in the intra-aggregate pores in the segmentation editor module (Fig. 2-C). After acquiring the mask, the intra-aggregate pores were selected via image thresholding (Fig. 2-D). Segmentation was performed based on the total grayscale histogram. First, two thresholds ($T_{\min}$ and $T_{\max}$) were selected based on visual observation of the grayscale histogram. The pixels with grayscale values lower than $T_{\min}$ and higher than $T_{\max}$ were converted to 0 and 1, respectively, while the pixels with grayscale values between $T_{\min}$ and $T_{\max}$ were the mixed voxels of pores and other substances. To eliminate these mixed voxels, the standard deviation (σ) between the average neighboring voxel value and the central voxel value was calculated. When $\sigma < 0.1$, the voxels were classified as low variability voxels (Schluter et al., 2010; Borges et al., 2019). The visualization of pore networks is displayed in Fig. 2-E, and the pore parameters were obtained by the computation module (Fig. 2-F). The total porosity, equivalent diameter, surface area, volume, pore number, and pore size distribution (categorized by pore volume and equivalent diameter) were obtained using Avizo 9.0 (FEI, 2016); the pore number density was defined as the ratio of the number of pores to the volume of the mask; and the pore shape factor (SF) was calculated with equation (1).

$$SF = \frac{A_0}{A} \quad (1)$$

where A_0 is the surface area of the equivalent sphere of the pores, and A is the actual surface area of the pores (Zhao et al., 2017; Zhou et al., 2012).

2.3. Statistical analyses

The data processing and statistical analyses in this study were

conducted using SPSS 13.0 software. The differences among the pore structure parameters of soil aggregates in the different soil types were analyzed by ANOVA after homogeneity of variance and normality tests. The Pearson correlation coefficient was used to examine the correlations between different pore parameters and between the organic matter content and the pore parameters.

3. Results

3.1. Visualization of intra-aggregate pore networks

Fig. 3 shows the 3D visualizations of the pore networks of soil aggregates for the three soil types. The spatial pore characteristics of the soil aggregates were significantly different among the three soil types. The total porosity of *chestnut* soil aggregates was the highest among the three soil types; the pores were mostly small and independent, with dense and random distributions (Fig. 3-A). *Red* soil aggregates had a low total porosity; the distribution of large pores and small pores was relatively dispersed (Fig. 3-B). The total porosity of the *alpine meadow* soil aggregates was higher than that of the *red* soil aggregates and lower than that of the *chestnut* soil aggregates; the pore distribution was relatively dense (Fig. 3-C).

Fig. 4 shows the pore network of the central part of the soil aggregates in 2D to more clearly visualize the internal structure of the aggregates. Several elongated pores were observed in the *alpine meadow* soil (Fig. 4-A). The soil pores in the *red* soil aggregates were mostly micropores and macropores (Fig. 4-B). The soil pores in the *chestnut* soil aggregates were small and dense and were more continuous than those of the *alpine meadow* soil and *red* soil; these characteristics may be attributed to the grass/shrub root system in the *chestnut* soil (Fig. 4-C).

3.2. CT-measured intra-aggregate pore characteristics

The soil pore characteristics of the soil aggregates are listed in Table 2. The total porosity of the *chestnut* soil aggregates was high

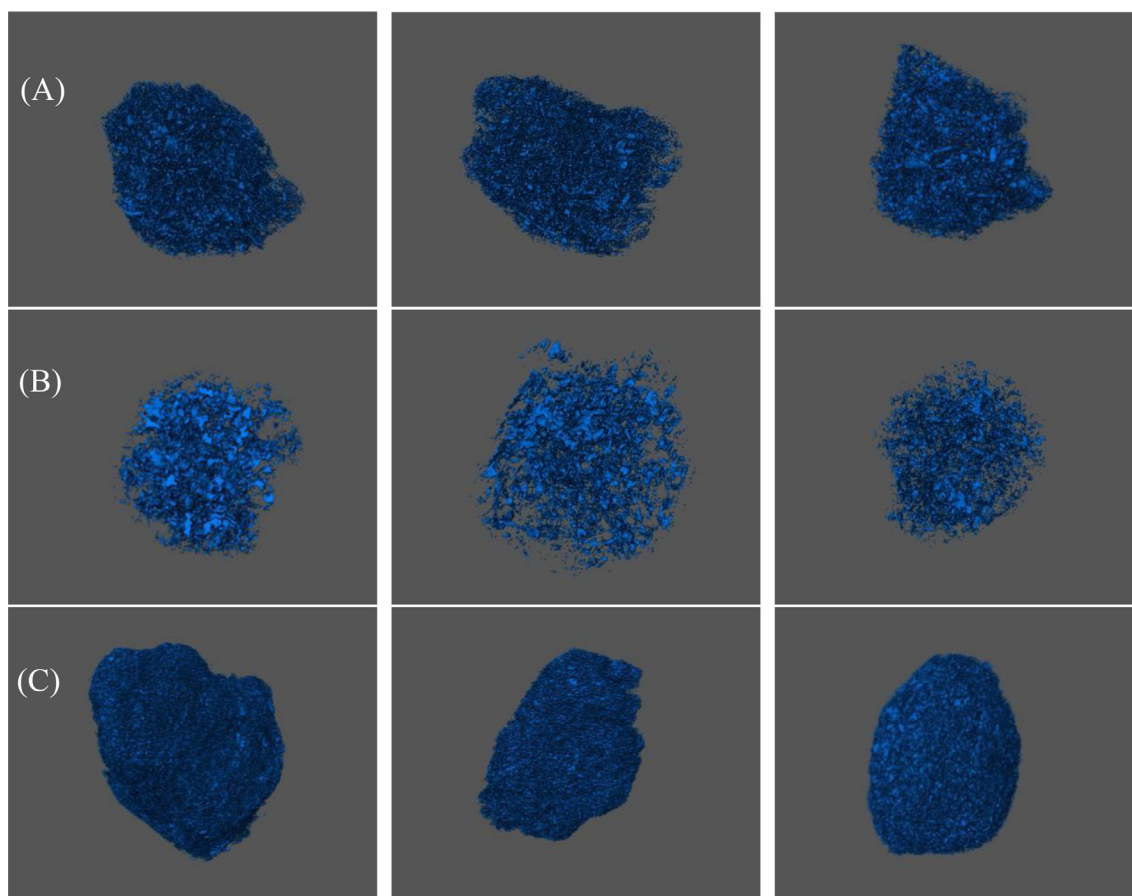


Fig. 3. 3D visualization of the pore networks in aggregates: (A) *alpine meadow* soil; (B) *red* soil; (C) *chestnut* soil.

($0.14 \mu\text{m}^3 \mu\text{m}^{-3}$) and was 2.06 times higher than that of the *alpine meadow* soil aggregates ($0.068 \mu\text{m}^3 \mu\text{m}^{-3}$) and 3.04 times higher than that of the *red* soil aggregates ($0.046 \mu\text{m}^3 \mu\text{m}^{-3}$). The *chestnut* soil aggregates had the highest pore number density ($18.33 \times 10^5 \text{ no.}\mu\text{m}^{-3}$), which was two times higher than that of the *alpine meadow* soil aggregates ($6.71 \times 10^5 \text{ no.}\mu\text{m}^{-3}$) and *red* soil aggregates ($5.78 \times 10^5 \text{ no.}\mu\text{m}^{-3}$). Significant differences in soil total porosity were found between the *chestnut* soil and the *red* and *alpine meadow* soils.

The *alpine meadow* soil aggregates had a large mean equivalent diameter of $23.907 \mu\text{m}$, a mean surface area of $9\,966 \mu\text{m}^2$ and a mean volume of $105\,533 \mu\text{m}^3$, whereas the *chestnut* soil had a low mean equivalent diameter of $16.470 \mu\text{m}$, a mean surface area of $1\,257 \mu\text{m}^2$ and a mean volume of $6\,622 \mu\text{m}^3$. There were significant differences in mean surface area between the *chestnut* soil and the *red* and *alpine meadow* soils.

Soil pore morphology parameters were used to describe some of the soil pore characteristics (Zhou et al., 2012). The soil pore shape factor is an index that reflects the shape of pores (Zhou et al., 2012; Zhao et al., 2017a). The mean pore shape factor of the *alpine meadow* soil aggregates (0.227) was smaller than that of the *red* soil aggregates (0.238) and *chestnut* soil aggregates (0.249; Table 2), which indicated that the soil pores in the *alpine meadow* soil aggregates were relatively irregular.

3.3. Intra-aggregate pore size distribution

The soil pore volume size distribution is shown in Fig. 5. The percentage of soil pores in each size class decreased with increasing pore volume size. The percentage of pores in the size range of $0\text{--}2\,000 \mu\text{m}^3$ accounted for 40–60% of the pore volume (*alpine meadow* soil 41.61%, *red* soil 59.68% and *chestnut* soil 62.65%), and the percentage of pores

with a volume above $6\,000 \mu\text{m}^3$ was never higher than 10% (*alpine meadow* soil 8.36%, *red* soil 5.47% and *chestnut* soil 1.82%). Therefore, soil pores in the size range of $0\text{--}2\,000 \mu\text{m}^3$ dominated the aggregates in terms of their number. The percentage of pores with a volume of $0\text{--}2\,000 \mu\text{m}^3$ in *chestnut* soil was the highest (62.65%), and the percentages of pores of $10\,000\text{--}30\,000 \mu\text{m}^3$, $30\,000\text{--}60\,000 \mu\text{m}^3$ and $> 60\,000 \mu\text{m}^3$ volume were low (8.75%, 2.62%, and 1.82%) in the *chestnut* soil. The percentages of pores with a volume of $2\,000\text{--}5\,000 \mu\text{m}^3$, $5\,000\text{--}10\,000 \mu\text{m}^3$, $10\,000\text{--}30\,000 \mu\text{m}^3$, $30\,000\text{--}60\,000 \mu\text{m}^3$ and $> 60\,000 \mu\text{m}^3$ in *alpine meadow* soil aggregates were higher than those in *chestnut* and *red* soil. Therefore, the number distribution of the soil pore volume size in *alpine meadow* soil aggregates was more uniform. Moreover, the percentage of pores with a volume of $> 60\,000 \mu\text{m}^3$ in *chestnut* soil (1.82%) was three times lower than that in *red* soil (8.36%) and two times lower than that in *alpine meadow* soil (5.47%).

Fig. 6 shows the equivalent diameter distribution of pores inside the aggregates. According to the standards of the American Soil Institute, the soil pores were classified into four kinds: macropores ($> 80 \mu\text{m}$), mesopores ($30\text{--}80 \mu\text{m}$), micropores ($5\text{--}30 \mu\text{m}$) and ultramicropores ($< 5 \mu\text{m}$) (Lal and Shukla, 2004). The percentage of pores in each classification decreased as the pore equivalent diameter increased. The percentage of pores with an equivalent diameter of $0\text{--}30 \mu\text{m}$ was $> 75\%$, and the percentage of pores with an equivalent diameter of $> 80 \mu\text{m}$ was $< 3\%$. Therefore, micropores dominated the aggregates in terms of their number; the percentage of micropores decreased in the following order: *chestnut* soil (90.39%) $>$ *red* soil (85.55%) $>$ *alpine meadow* soil (75.67%).

3.4. The shape of the intra-aggregate pores

The pore shape distribution in the soil aggregates is shown in Fig. 7.

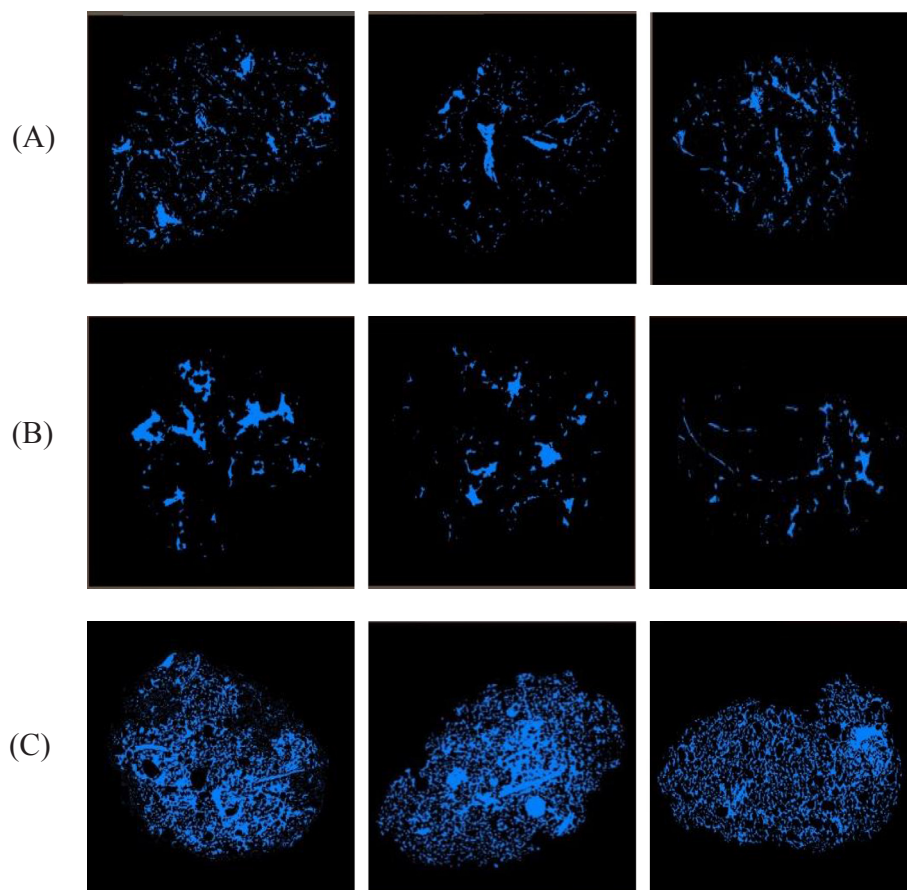


Fig. 4. 2D visualization of the pore networks in aggregates: (A) *alpine meadow* soil; (B) *red* soil; (C) *chestnut* soil.

The closer the pore shape factor is to 0, the more irregular and elongated the pores are. The closer the pore shape factor is to 1, the more regular and spherical the pores are. The percentage of pores with a pore shape factor between 0 and 0.1 in *red* soil and *alpine meadow* soil aggregates was equal (1.42%) and was larger than that in *chestnut* soil aggregates (0.81%). Therefore, there were more elongated pores in the *red* soil and *alpine meadow* soil aggregates than in the *chestnut* soil aggregates. The concentrated distribution values for the pore shape factor in *red* soil, *alpine meadow* soil and *chestnut* soil were 0.1–0.3 (67.04%), 0.1–0.3 (74.10%) and 0.2–0.4 (74.23%), respectively. Additionally, the percentages of pores with a pore shape factor between 0.3 and 0.4 in *red* soil, *alpine meadow* soil and *chestnut* soil aggregates were 31.53%, 24.48% and 34.59%, respectively. Therefore, the pores inside the *chestnut* soil aggregates were more regular than those in the *red* soil and the *alpine meadow* soil aggregates.

4. Discussion

The results showed that the pore characteristics of the soil

aggregates of the three soil types varied greatly. The pores inside the *chestnut* soil aggregates were mostly small and independent, with a dense and random distribution; this may be attributed to the root system of the grass/shrubs in the soil origin area. This is consistent with the findings of Hu et al. (2015) indicating that a large number of macropores exist in soils developed under shrub vegetation, which may be due to the well-developed shrub root system and its subsequent decay (Udawatta and Anderson, 2008; Kravchenko et al., 2011; Zhao et al., 2017b; Hu et al., 2018; Hu et al., 2019). Moreover, the pores of the *alpine meadow* soil were more random, irregular and elongated, which can be attributed to the freeze–thaw cycles in the alpine region. *Alpine meadow* soils develop in high-latitude areas under unique and harsh climate conditions that are distinguished by low temperatures, ultraviolet radiation, freeze–thaw cycles and short growing seasons. Luo et al. (2010) found that macropores developed by freeze–thaw or wet-dry cycles were randomly and less continuously distributed. The *red* soil had a low porosity and poor soil structure. This may be related to aspects of its original environment, such as high temperatures and abundant precipitation, and its intrinsic soil properties (e.g., low soil

Table 2

Pore characteristics of aggregates measured using computed tomography (CT).

Soil Sample	Total porosity ($\mu\text{m}^3 \mu\text{m}^{-3}$)	Mean equivalent diameter (μm)	Mean surface area (μm^2)	Mean volume (μm^3)	Pore number density ($10^5 \text{ no.} \mu\text{m}^{-3}$)	Mean pore shape factor
<i>Alpine meadow</i> soil	$0.068 \pm 0.018(\text{b})$	23.907 ± 0.446	$9966 \pm 2068(\text{a})$	105533 ± 32361	6.71 ± 1.74	0.227 ± 0.003
<i>Red</i> soil	$0.046 \pm 0.008(\text{b})$	19.172 ± 0.449	$7349 \pm 1233(\text{a})$	83366 ± 24357	5.78 ± 1.11	0.238 ± 0.003
<i>Chestnut</i> soil	$0.140 \pm 0.023(\text{a})$	16.470 ± 2.820	$1527 \pm 631(\text{b})$	6622 ± 3297	18.33 ± 10.26	0.249 ± 0.014

Column for each data group by the same letter are not significantly different at $p < 0.05$, the absence of letters indicates no significant differences for that property.

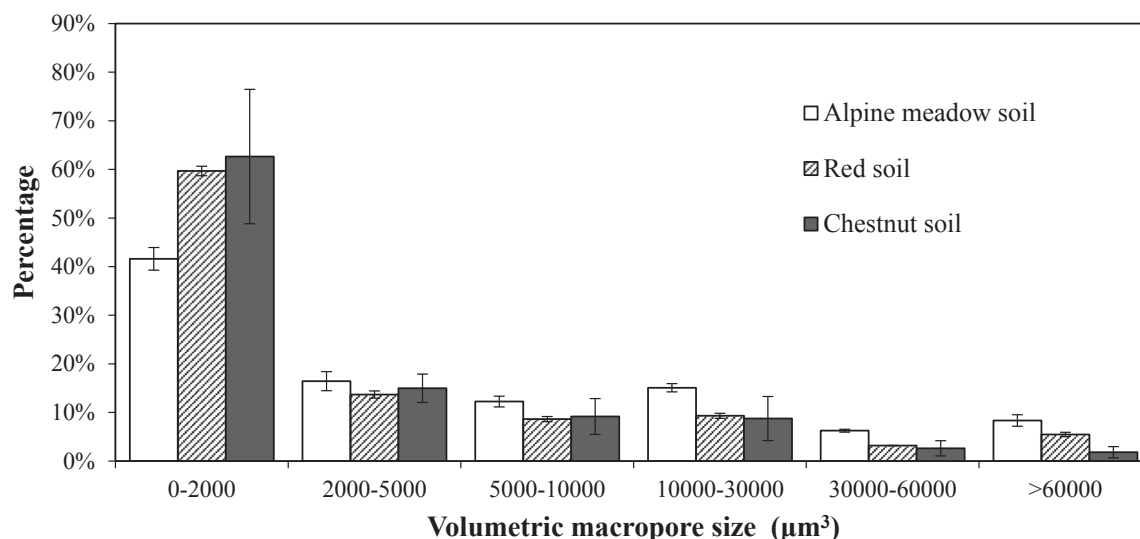


Fig. 5. Pore size distribution (by volume) of soil aggregates of three soil types.

pH, poorly developed soil structure, low cation exchange capacity, and low nutrient availability) (Xu et al., 2003; Liu et al., 2014). Moreover, previous studies have indicated that the formation and stability of pores are likely controlled by intrinsic soil properties (e.g., particle size distribution and mineralogy) (Lamandé et al., 2011).

Some studies have shown that high clay and silt contents result in a low number of pores and that pore density increases with sand content (Park and Smucker, 2005). This is consistent with the results of our study. The results showed that the total porosity of the chestnut soil ($14 \mu\text{m}^3 \mu\text{m}^{-3}$) was the highest among the three soil types; the chestnut soil had relatively low contents of clay and silt (12% and 11.88%, Table 1). In the red soil, the total porosity was the lowest ($0.008 \mu\text{m}^3 \mu\text{m}^{-3}$), which may be related to its higher clay content (43.88%, Table 1). This is consistent with the results of other studies. Lugato et al. (2009) indicated that the soil aggregate porosity (range $0.007\text{--}100 \mu\text{m}$) was higher ($P < 0.01$) in sandy soils than in clayey soils. Henderson et al. (2005) found that soils with higher sand fractions tend to have more macroporosity than clay soils. Moreover, a strong correlation between soil texture and soil porosity can be observed in Table 3. The results showed that there was a significant positive correlation between the total porosity and the sand content. Significant negative correlations between the soil total porosity and the silt and clay contents were

also observed. Therefore, soil texture (sand, silt and clay) significantly influences soil aggregate total porosity. However, Lugato et al. (2009) found a strong negative correlation between clay content and the proportion of pores $5\text{--}100 \mu\text{m}$, while clay content was positively correlated with the proportion of pores $< 0.1 \mu\text{m}$. This study could only analyze pores $> 7 \mu\text{m}$ in diameter. Therefore, other methods, such as mercury porosimetry, should be used to study smaller pores.

SOM is another important factor that affects soil structure and is a key factor in the formation of aggregates and soil C processes (Denef and Six, 2005; Zhao et al., 2010; Kravchenko et al., 2014; Piccoli et al., 2019). Significant positive correlations between the soil total porosity and SOM were observed (Table 3). This is consistent with previous results showing that SOM increased the soil porosity of aggregates (Nakano et al., 2015; Zhao et al., 2017b). De Jonge et al. (2009) found that sponge-like pore networks were formed in some soils, while other soils formed pipe-like structures; these structures might be related to the SOM content. Although there was no correlation between SOM and the mean pore shape factor (Table 3), strong correlations between SOM and pore shape factors of $0.1\text{--}0.2$, $0.2\text{--}0.3$ and $0.3\text{--}0.4$ were observed (Table 4). There were significant and positive correlations between the pore shape factors of $0.1\text{--}0.2$ and $0.2\text{--}0.3$ and SOM, and SOM was negatively correlated with the pore shape factor of $0.3\text{--}0.4$ (Table 4).

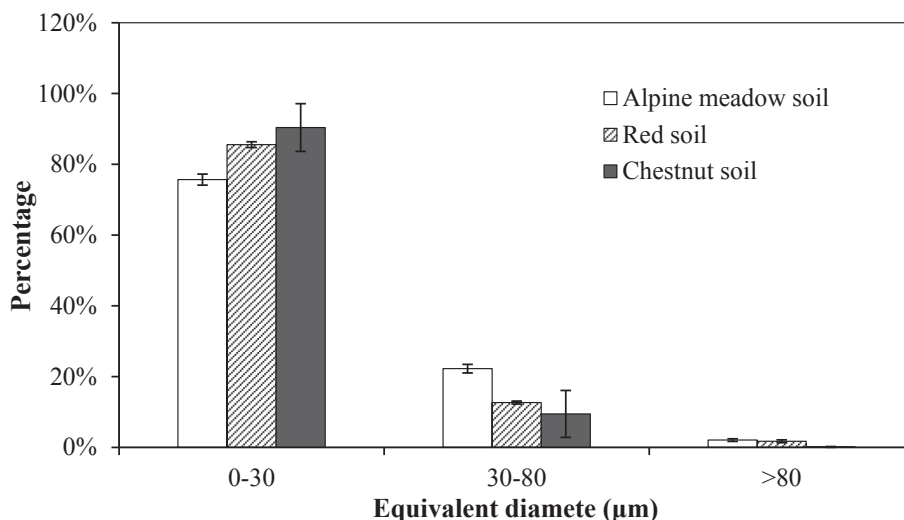


Fig. 6. Pore size distribution (by equivalent diameter) of soil aggregates of the three soil types.

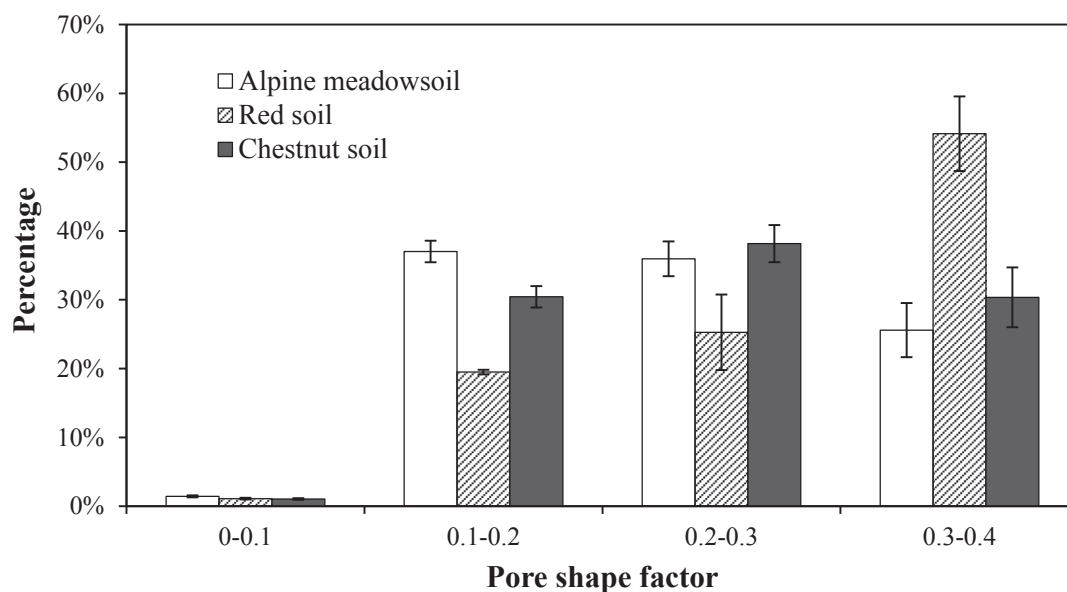


Fig. 7. Pore shape distribution of soil aggregates of the three soil types.

Table 3

Correlations between the pore parameters and selected physico-chemical characteristics.

	SOM	Sand	Silt	Clay
Total porosity	0.671*	0.898**	−0.911**	−0.856**
Mean volume	−0.275	−0.696*	0.834**	0.603
Mean pore shape factor	−0.377	−0.778*	0.900**	0.688*
Pore number density	0.476	0.748*	−0.822**	−0.683*
Pd < 30	−0.108	0.357	−0.608	−0.223
Pd ₃₀₋₈₀	0.174	−0.282	0.540	0.148
Pd > 80	−0.375	−0.789*	0.924**	0.693*

Pd < 30, Pd₃₀₋₈₀ and Pd > 80 represent pores with pore diameter < 30 μm, 30–80 μm and > 80 μm, respectively;

*Correlation is significant at the 0.05 level.

**Correlation is significant at the 0.01 level.

Table 4

Correlations between the pore shape factor (PSF) and soil organic matter.

	PFS (0–0.1)	PFS (0.1–0.2)	PFS (0.2–0.3)	PFS (0.3–0.4)
SOM	0.287	0.900**	0.850**	−0.928**

*Correlation is significant at the 0.05 level.

**Correlation is significant at the 0.01 level.

Therefore, SOM may result in more irregular pores and fewer regular pores. This is consistent with the results of Zhao et al. (2017b), which indicated that SOM was positively correlated with the fraction of elongated pores but negatively correlated with the fractions of regular and irregular pores.

Although macropores make up only a small part of the soil (Jarvis, 2007), air and water can flow preferentially through them. Thus, they are critical in the transport of water and dissolved constituents to the unsaturated zone. Mesopores provide good air permeability and rapid water flow that can be consistently used by plants (Castellini and Ventrella 2012; Liu et al., 2009). Moreover, soil mesopores have good connectivity (Schwen et al., 2011). Soil micropores are unable to provide water either quickly or continuously enough to meet the needs of crop growth or to allow water to be taken up by crop roots (Liu et al., 2009). The presence of smaller pores in soil is related to greater water stability (Dal Ferro et al., 2012). There were relatively more meso- and macropores in *alpine meadow* soil aggregates than in the other soil

aggregates, and these pores were irregular and elongated with sufficient connectivity and air permeability. Therefore, *alpine meadow* soil aggregates would be favorable for the transportation and storage of oxygen and water. This is consistent with the study by Zhao et al. (2017b), which found that elongated pores were important for the transport and storage of water and gas. The total porosity of *chestnut* soil aggregates was the highest among the three soil types. However, the micropores in the *chestnut* soil aggregates were highly discontinuous, which would cause soil water to move slowly and discontinuously. *Red* soil is generally low in organic carbon, lacks available nutrients and demonstrates inherent fragility, in addition to having a poor soil structure (Xu et al., 2003; Zhang and Xu, 2005; Basha et al., 2016; Jin et al., 2019). This is consistent with our results, which indicated that *red* soil aggregates had the lowest total porosity and low organic matter content. It is speculated that the *alpine meadow* soil aggregates may be favorable for the transportation and storage of water and oxygen; that water in the *chestnut* soil aggregates moves slowly and is thus mostly unusable for plants; and that the *red* soil structure is poor and susceptible to rapid erosion.

This study highlights the usefulness of X-ray CT for quantifying the pore structure of soil aggregates and the importance of aggregate pore structure for different soil groups. These factors could be potential parameters for future soil modeling. However, the application of X-ray CT is limited because only part of the total porosity is quantified due to the image resolution. In addition, only three to five aggregates were selected from each treatment for CT scanning due to the high scanning cost (Papadopoulos et al., 2009; Zhou et al., 2012; Wang et al., 2012; Zhou et al., 2013), which creates uncertainty in the interpretation of these results. Therefore, future studies need to increase the number of aggregate samples assessed, and the CT method should be combined with other traditional methods to verify the accuracy of the results.

5. Conclusions

The findings of the study verified our hypothesis that intra-aggregate pore structures varied greatly among different soil groups and that every soil group had its own pore characteristics controlled by different biotic or abiotic factors. In the *chestnut* soil aggregates, the pores were mostly small and densely distributed, which may be related to the presence of grass/shrub roots. *Alpine meadow* soil aggregates were characterized by many elongated pores and macropores, which may be attributed to the freeze–thaw cycles in the alpine region. *Red*

soil aggregates had lower total porosity and less pore connectivity than the other soil aggregates, which is likely related to the high temperatures and abundant precipitation in the subtropical region. The study also showed that soil texture and SOM significantly influence the total porosity of the soil aggregates. There were significant positive correlations between the soil total porosity and the sand content and significant negative correlations between the soil total porosity and the silt and clay contents. Furthermore, higher SOM may result in a higher number of irregular pores in aggregates.

Conflict of interest statement

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and there is no professional or other personal interest of any nature or kind in any product, service or company that could be construed as influencing the position presented in the manuscript.

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